Flood Watch Punjab:
satellite flood detection and reservoir-level forecasting

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Abstract

Effective and accurate flood detection and warning systems require timely inundation detection with a reliable and constantly available source of information on reservoir conditions. This study aims to assess floods in the region of Punjab and provide an accurate flood mapping using Sentinel-1 Synthetic Aperture Radar(SAR) observations of Punjab processed through a custom pipeline to refine classifications and also determine whether combining satellite prediction signals with river catchment derived signals improved inundation prediction. Independently evaluated fixed reduction of Copernicus Global Flood Monitoring(GFM) data for 52 locations using SAR and Landsat-9 observations of a total of 673.95 km^2 of usable coverage out of a total of 1,577.16km^2 of total coverage showed that, among the locations with conclusive image based references, the sampling-weighted precision and recall were 93.8% and 66.1%, respectively. Reservoir forecasts for Bhakra and Pong reservoirs were assessed chronologically through 2021-24 and tested against persistence and a matching undamped pattern. The river reservoir prediction showed for Bhakra, the MAE (mean absolute error) value was 0.247m meanwhile for Pong it was 0.916m. Bhakra improved over persistence for all 5 horizons showing a 36.6% MAE reduction over the 5th horizon. The encountered bottlenecks which led to difficulties and a higher MAE include but are not constrained by the limited access to real-time reservoir and satellite data. Early and accurate prediction along with a system of rapid response to predictions would save structures and prevent livestock, ,monetary and agricultural damages.

Introduction

In Punjab, there is satellite imagery that can show inundated land, there are reservoir records which describe stored water, and there are weather forecasts that estimate future rainfall.Rainfall and reservoir water levels can influence downstream flooding in the Sutlej basin [2] However, connecting these sources is only useful if the resulting warning can be accurate and consistent.The primary question is how accurately can we identify and detect inundation in areas of Punjab using Sentinel-1 GFM classifications in comparison to near-time image references(optical)[1][3].And also, our secondary question is how accurately we can predict reservoir levels and how useful is the amalgamation of the prediction and detection is in overall flood prediction.

Innovation

The key innovation of this study is the amalgamated evaluation framework made, keeping in mind the Punjab region. It combines flood mapping and analysis from satellites with a forecast of river reservoir levels. It is also designed with the constraints in mind that come with each of the data source due to the unpredictable nature of weather and satellite imagery. The custom pipeline filters the pre existing GFM quality and metrics while using contextual spatial layers. It also treats unverified regions as unclassified and uses the available region to calculate the net surface area coverage. The Independent image verification evaluation combined with the signal and prediction experiments helps us with the identification on which outputs are supported by evidence(evaluated).

Methodology

The Sentinel-1 deriver Copernicus GFM classifications two masks: i) Quality, ii) Contextual. Flood area along with classified area and flood fraction were calculate using the following equation for each satellite observation captured at a particular time :

$$F = \sum_p a_p v_p f_p, \qquad V = \sum_p a_p v_p,$$
$$q = \frac{F}{V} \quad (V > 0).$$

Here, a_p is the intersecting pixel area, v_p shows usable evidence and f_p shows mapped flood. Unsupported pixels here however, remained unknown. The test comparison used 673.95 km^2 of usable evidence out of a 1,577.16 km^2 pilot. The Sentinel-1 and Landsat-9 observations were separated by a mere 4.62 hours, with the earlier Sentinel-2 imagery providing context. Fifty-two locations were reviewed. Three uncertain references were excluded; precision, recall, F1 and IoU used raw counts and inverse-probability sampling weights were calculated using the following equation:

$$w_i = \frac{1}{p_i},$$

where p_i is the sample inclusion probability.

The river reservoir prediction uses historical reservoir levels to estimate recent trends and damped forecasts:

$$g_{t,k} = \frac{z_{t-1} - z_{t-1-k}}{k},$$
$$\widehat{z}_{t+H} = z_{t-1} + \alpha_{k,H}(H+1)g_{t,k}.$$

Here, z is reservoir level, k is lookback and H is nominal horizon. Damping coefficients were fitted by constrained least squares within $[0, 1]$. Persistence held the latest level; the undamped comparator used $\alpha = 1$. Comparisons used identical eligible cases; missing histories and

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} \left|\widehat{z}_i - z_i\right|.$$

The integration experiment used median imputation, standardization and balanced, L2-regularized logistic regression:

$$P(y = 1 \mid x) = \frac{1}{1 + \exp[-(\beta_0 + \beta^\top x)]}$$

Here, x contains the selected predictors and β_0 is the intercept, and $\beta^\top x$ is the weighted sum of the input features .

The satellite, river and combined predictors used 19 cases before 2020 and 20 cases from the past year, River features for prediction included rainfall forecasts and reports. The threshold for inundation remained as 0.5 (prediction probability).

Results

Among 49 decided image references, the custom satellite pipeline recorded eight correct flood detections, two misses, two false alarms and 37 correct non-flood classifications. Raw precision and recall were 80.0%, with F1 0.800 and IoU 0.667; sampling-weighted precision and recall were 93.8% and 66.1%.

At Bhakra's nominal first horizon, MAE was 0.247 m versus persistence's 0.488 m across 194 paired cases, a 49.3% reduction. At the fifth horizon, MAE was 0.761 m versus 1.201 m across 167 cases, a 36.6% reduction. Pong's first-horizon MAE was 0.916 m across 12 cases; its undamped comparator performed better at horizons 2–5. All three integration predictors missed ten NRSC mapped-positive test cases, including after reference re-review. These results establish imagery agreement and component forecasting findings, without demonstrating improved integrated flood warnings.

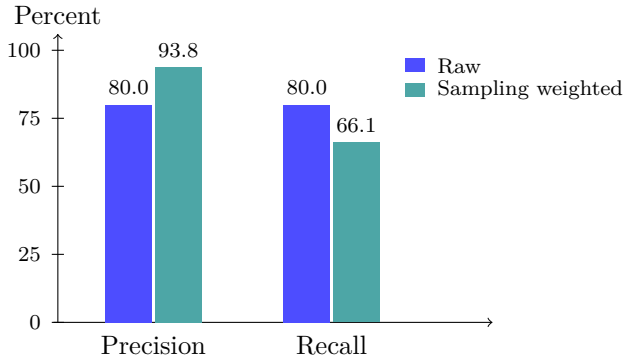


Figure 1: Raw and sampling-weighted image-reference precision and recall.

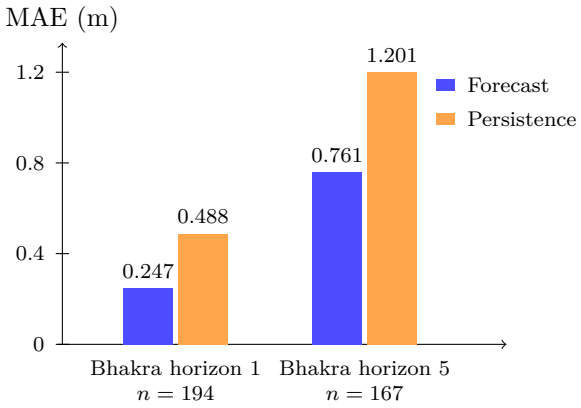


Figure 2: Bhakra reservoir-level mean absolute error at two reported horizons.

Acknowledgements and References

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[2] Amin et al. (2026). Transboundary flood dynamics in the Sutlej basin (2025): Sentinel-1 SAR

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[3] García-Pintado et al. (2015). Satellite-supported flood forecasting in river networks: A real case study. Journal of Hydrology, 523, 706–724. https://doi.org/10.1016/j.jhydrol.2015.01.084